

ANALYZING EVIDENCE -- A2 Requirements and Scope Specification Source:
Requirements3.pdf (formal FR/BR/US spec) Stage: Analyzing
??

1 What this document captures

The requirements PDF was written after the agency research (A1) and before any code. It translates the assignment brief into a formal specification: functional requirements, business rules, user stories, and acceptance criteria. Ten FRs were drafted; only five were built in the POC.

1 Functional Requirements -- full list with POC status

FR1 Input validation Scope: reject non-image files, enforce ?16MB limit, require non-empty description Status: [x] BUILT -- app.py lines 33-43, allowed_file()

FR2 Vision analysis and tagging Scope: product_type, ?3 product_tags, mood, target_signals, suggested_hashtags extracted from image Status: [x] BUILT -- agents/vision.py, OpenRouter vision models

FR3 Theme selection Scope: user picks from 3 campaign themes before copy generates; each theme changes tone, color palette reference, hashtags Status: [] DEFERRED -- added complexity without clear output value over the tone slider approach

FR4 Campaign copy generation Scope: exactly 2 variants, each ?150 chars primary text, distinct creative angles, clear CTA in each variant Status: [x] BUILT -- agents/copywriter.py
Note: variant angle constraint (emotional/benefit) added in Realizing phase after observing generic output

FR5 Audience persona Scope: persona_label, age_range, interests (list), platform_behavior Status: [x] BUILT -- agents/audience.py

FR6 Landing page skeleton Scope: generate a minimal HTML landing page from campaign output; hero copy, CTA button, color palette applied Status: [] DEFERRED -- useful but not core to the campaign proof of concept; output is already professional

FR7 Compliance guardrails Scope: flag guarantee claims, health claims, superlatives, urgency language, scientific claims, weight loss claims; return flags with variant ID, matched text, reason Status: [x] BUILT -- agents/guardrails.py (7 regex patterns)

FR8 Structured output Scope: JSON output with all required keys; human-readable summary in addition to raw JSON Status: [x] BUILT -- app.py output dict, result.html rendering

FR9 Regeneration with different parameters Scope: user selects different tone or channel, full pipeline reruns Status: [] DEFERRED but EVOLVED -- became the interactive ReAct refinement loop (richer than a simple re-run) See DL-09 for the design evolution

FR10 Evaluation scoring Scope: compare variant quality using a pre-defined lexicon; score each variant on clarity, audience fit, urgency Status: [] DEFERRED -- requires a reference lexicon that does not exist yet; CTA scoring (FR8) partially covers this

1 Business Requirements

BR1 Policy-safe output No fabricated prices, invented specifications, or unverifiable claims. Guardrails (FR7) addresses the most common violation patterns. The vision agent is explicitly instructed not to invent features not visible in the image or description.

BR2 Thematic consistency Color palette extracted from the product image must be referenced in copy generation context. The dominant color is passed to the copywriter so color references in copy match the actual product.

BR3 No fabricated statistics Copy must not include percentage claims, survey data, or clinical results that were not present in the input. Guardrails pattern "scientific claim" catches "clinically proven" and similar. The vision agent prompt explicitly says "do not invent features."

BR4 JSON + text output Every pipeline run produces a structured JSON output (for export) and a human-readable HTML report (result.html). Both update live during the refinement loop.

1 User Stories

US-1 As a small business owner, I want to upload a product photo and short description and receive two ready-to-post ad copy variants, so I can launch a campaign without hiring an agency. Acceptance: ?2 minutes from upload to visible output.

US-2 As a marketing student, I want to see which agents ran and why, so I can understand how the multi-agent pipeline works. Acceptance: routing label visible in refinement panel.

US-3 As a compliance reviewer, I want flagged risky claims highlighted, so I can decide whether to edit before publishing. Acceptance: compliance flags shown with matched text and reason.

US-4 As a product manager, I want to refine the copy by tone and audience age without re-running the full pipeline, so I can iterate quickly in a live meeting. Acceptance: slider changes + Refine button triggers only the affected agents, not the vision analysis.

US-5 As a developer evaluating the system, I want to export the full campaign JSON with one click, so I can integrate the output into other tools. Acceptance: /export/ returns valid JSON with all keys.

1 What the requirements revealed

Compliance is not optional. FR7 was marked as a required functional requirement -- not a nice-to-have. The brief explicitly says "policy-safe output" (BR1). This made guardrails a mandatory stage in the pipeline, not an optional quality gate. That had architectural consequences: see DL-06.

FR9 evolved. The original requirement imagined a simple "regenerate with different theme" button. During Realizing, the refinement loop (ReAct routing + slider controls + proactive suggestions) became something more capable and more interesting. The FR9 intent is fully delivered; the implementation is richer than the spec anticipated.

FR3 was the right deferral. Theme selection would have required a reference system for what each theme implies across tone, hashtag style, and color references. Building that before the base pipeline was stable would have been premature. The tone slider covers the core use case.

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